

Use Case Add Product

1. Use case code

UC234.1

2. Brief Description

This use case allows the Product Manager to add a new media product (Book, Newspaper, CD, DVD) to the AIMS system. The system will check data format constraints and pricing rules before officially saving it to the database.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally.

5. Basic Flow of Events

1. Product Manager select Add Product
2. Software displays general form for Product Manager
3. Product Manager input general information and choose specific type of product.
4. Software displays form for fill specific information depending on product type
5. Product Manager input specific information and submit.
6. Software checks the data and price rule.

6. Alternative flows

No	Location	Condition	Action	Resume location
1	At step 6	If the input data is not in the correct format	Software display detailed error for each data field	At step 3 or 5

2	Any step	If Product Manager cancel process	Software do not save any changes and return product list interface	End the usecase
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7. Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1	Barcode	Unique identifier for the product	Yes	Unique in the system; numeric only	893123456789
2	Title	The name of the media product	Yes	Cannot be blank	DVD Hello World
3	Category	Product classification	Yes	One of: Books, Newspapers, CDs, DVDs	Books
4	General Description	Condition, primary color, or return condition	Yes	Text description	New condition, blue cover
5	Original Value	Base price excluding 10% VAT	Yes	Positive integer	100
6	Current Price	Selling price excluding 10% VAT	Yes	Between 30% and 150% of Original Value	120
7	Dimensions	Physical size (Height x Width x Length)	Yes	Positive values only	20 x 15 x 3 (cm)
8	Weight	Physical weight of the product	Yes	Positive value (kg)	0.5
9	Author	Creator(s) of the book	Yes (Book)	Required if Category is "Books"	Ngo Tat To
10	Cover Type	Format of the book cover	Yes (Book)	"Paperback" or "Hardcover"	Paperback
11	Editor-in-chief	Head of the newspaper	Yes (News)	Required if Category is "Newspapers"	Nguyen Van A
12	Artists	Performers or singers	Yes (CD)	Required if Category is "CDs"	Son Tung M-TP

13	Track List	List of songs with titles and lengths	Yes (CD)	Must include title and duration per track	Track 1 - 10:30
14	Disc Type	Physical format of the disc	Yes (DVD)	"Blu-ray" or "HD-DVD"	Blu-ray
15	Director	Director of the media	Yes (DVD)	Required if Category is "DVDs"	Christopher Nolan

8. Output data

No	Data fields	Description	Display format	Example
1	Operation Status	Notification of the result for the performed action (add, edit, delete, or adjust).	Text notification	Product added successfully
6	Error Message	Detailed notification when input data is invalid or violates system rules.	Text (Red)	Price must be between 30% and 150% of Original Value

9. Postconditions

- Stored data: All products successfully saved into database or new information be updated with newest version.
- Error message: Product Manager receive error text message at invalid field.